

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

When this project started, the aim was simple: help people report an emergency easily, keep that information organised, and protect it properly. At the end of the work, that aim has been achieved. RakshaSafe is a working web application. A user can register, raise an SOS incident, share the location where possible, and see what is happening with that incident through notifications and history. An administrator on the same platform can watch all the incidents, change their status step by step, assign rescue teams, review unsafe-area reports, and make reports from the real data stored in the system.

The project was done step by step, and each step used the previous one. The requirements and analysis stage decided what the system must do, and also what it must not promise. Then the design stage made the plan: a React frontend, a Node.js backend, a separate Python AI service for risk assessment, and MongoDB for storing data. Next came the implementation, module by module, and finally testing, which checked everything against the requirements — functions, security, data safety, validation, and usability. Following the documentation so closely is itself a big achievement of this project.

Three results really matter. First, the incident record is clean and complete: type, category, description, priority, status, location when available, and a full history of every change. Second, the system is honest: if location is not available, it never creates a fake one, and if an SMS or email provider fails, the notification is marked as failed instead of breaking the incident. Third, the AI risk service is kept separate and straightforward, so it gives explanatory support without making dangerous promises.

Security was also taken seriously. Passwords are hashed, login uses secure tokens, user and admin rights are clearly separate, and every admin action is recorded in logs. For a system that keeps personal details, locations, and emergency contacts, this is not a bonus — it is a must.

So, in short: RakshaSafe does what it was made to do. It reports and tracks emergencies, keeps resources and history in one place, protects sensitive data, and gives administrators a clear, truthful view of everything happening. The plan was honest, the design followed the plan, and the tested system behaves exactly as documented. The project is complete, working, and ready for the improvements described next.

7.2 Limitations

It is also fair to say what the system cannot do. RakshaSafe stores and organises information as software; it cannot guarantee that police, ambulance, or rescue teams will reach a place on time. It does not file police complaints automatically, and it does not give exact arrival times. The application needs the internet to work. It depends on the hosting setup, the database, the network, and any external services. Notifications only reach people if the email, SMS, or WhatsApp provider is set up and working. Location works only when the device gives permission and has good accuracy. And the AI risk result is only guidance, not a sure

prediction. These points are not faults; they are the honest limits inside which the system was built, and saying them clearly makes the rest of the work believable.

Technically, the risk model today works on fixed factor inputs rather than learning from past incidents. Notifications are fully recorded, but they rely on outside providers instead of built-in messaging. Maps and reports depend on external services, and the system runs as a web application, not a native mobile app. Every one of these points is a natural place where the project can grow.

7.3 Future Scope

RakshaSafe was built in modules on purpose, so the ideas below can be added without rebuilding everything. The main improvements are:

1. **1. Native mobile apps (Android & iOS):** A mobile app would make raising an SOS faster than a browser page. It could share location in the background and send push notifications, which helps a lot in real emergencies.
2. **2. Offline support:** If the internet is down, the app could store the incident on the phone first and sync it later. This is very useful in remote or disaster-hit areas where networks fail.
3. **3. Smarter AI with machine learning:** Instead of fixed rules, the risk service could learn patterns from past incidents, unsafe-area reports, and weather data. It would still stay honest — results would always be guidance, never a guarantee.
4. **4. Live police, ambulance and fire integrations:** Connecting to real response systems could escalate an incident automatically and let the police or ambulance team receive the report directly.
5. **5. Real-time hospital and shelter availability:** If connected to a reliable source, the system could show live bed availability and open shelters, so resource records become live status instead of static lists.
6. **6. Automatic or assisted FIR filing:** The system could help users fill formal police complaint forms quickly after an incident, reducing paperwork and delays.
7. **7. Voice-based SOS:** Speaking the emergency instead of typing it would be a big help in a panic situation, when a person may not be able to type.
8. **8. More languages and translation:** Adding more local languages and automatic translation would let more people across different regions use the system comfortably.
9. **9. Better analytics and predictions:** Trend charts, incident heat-maps, and scheduled reports would give administrators a stronger planning tool, all based on the data the system already stores.
10. **10. Multi-organisation support:** One installation serving many campuses or districts, with each organisation seeing only its own data, would make the system cost-effective to run at scale.

In the long run, RakshaSafe can grow from a single-organisation safety tool into a shared public safety platform, as long as it keeps the principles that made it trustworthy today: clear records, honest scope, protected data, and support that never overclaims.

As it stands now, the project is complete and tested. The documentation matches the implementation, the limitations are clear, and the future plans are practical rather than just imaginary. That is a solid and honest place for a final-year project to stand.